

Saturn

Overview

Saturn is the sixth planet from the Sun and the second largest in the Solar System, with an equatorial diameter of 116,460 km — about 9.5 times that of Earth. It is a gas giant composed mostly of hydrogen and helium, with no solid surface. Saturn is the least dense planet in the Solar System, with a mean density of 0.687 g/cm³ — less than the density of liquid water (1 g/cm³) — meaning that if a body of water large enough existed, Saturn would float in it. Saturn's rapid rotation (one day is 10.7 hours) combined with its low density gives it the most oblate shape of any planet; its equatorial diameter is about 10 % larger than its polar diameter. It orbits the Sun at an average distance of 1.43 billion km, taking 29.5 Earth years to complete one orbit.

Ring System

Saturn's ring system is the most spectacular and extensive in the Solar System, visible even through a small telescope from Earth. The rings span up to 282,000 km in diameter — more than two-thirds the distance from Earth to the Moon — yet are remarkably thin, averaging only about 10–20 metres in the densest regions, making them proportionally thinner than a sheet of paper. The rings are composed primarily of water-ice particles ranging in size from tiny dust grains to chunks several metres across, along with traces of rocky debris. They are divided into named groups (D, C, B, A, F, G, E) separated by gaps, the most prominent being the Cassini Division — a 4,800 km gap between the A and B rings caused by orbital resonance with the moon Mimas. Cassini mission data revealed that the rings are geologically young, probably between 10 and 100 million years old, meaning they formed long after the dinosaurs appeared on Earth, likely from the disruption of a moon or comet that ventured too close to Saturn. The rings are gradually raining material into Saturn's atmosphere and are expected to disappear completely within 100 to 300 million years.

Atmosphere and Polar Hexagon

Saturn's atmosphere, like Jupiter's, shows banding of light zones and dark belts, though the contrast is softer due to a haze of ammonia crystals in the upper atmosphere. Wind speeds on Saturn are among the fastest in the Solar System, reaching up to 1,800 km/h at the equatorial jet stream. One of the most extraordinary features of Saturn is the persistent hexagonal polar vortex at its north pole — a hexagon-shaped jet stream roughly 30,000 km across, first observed by Voyager 1 in 1980 and still active today. Each side of the hexagon is approximately 13,800 km long, larger than Earth's diameter. Periodically, Saturn experiences giant storms called Great White Spots that erupt from the equatorial or temperate regions, grow to encircle the entire planet, and then fade; one such storm in 2010–2011 was the largest observed in 20 years.

Titan and Enceladus

Titan is Saturn's largest moon and the second largest in the Solar System, with a diameter of 5,149 km — larger than the planet Mercury. It is the only moon in the Solar System known to have a dense atmosphere, which is primarily nitrogen (about 95 %) with methane clouds and a surface pressure 1.5 times that of Earth. Titan's surface temperature is -179 °C, cold enough for methane to exist as a liquid; rivers, lakes, and seas of liquid methane and ethane have been confirmed near its north pole by the Cassini spacecraft and the Huygens lander. Enceladus, Saturn's sixth-largest moon, has become one of the most exciting objects in astrobiology: Cassini discovered that it vents plumes of water vapour and ice particles from fractures near its south pole, fed by a global subsurface ocean of liquid water. The plumes contain hydrogen, silica particles, and complex organic molecules, suggesting hydrothermal activity at the ocean floor — conditions potentially suitable for microbial life. NASA's Dragonfly mission, a rotorcraft lander, is scheduled to reach Titan around 2034 to explore its prebiotic chemistry.